

ALL-MorphNet: A Novel Hybrid Deep Learning Architecture for Automated Acute Lymphoblastic Leukemia Diagnosis in Microscopic Images

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Abstract - Acute lymphoblastic leukaemia (ALL) is the most common cancer in children, but early detection is difficult because its symptoms are often subtle and easy to miss. While automated diagnostic tools hold significant promise, they face numerous challenges. These include a lack of diverse data, unbalanced datasets, and the requirement for high-performance computing. To address these challenges, we developed ALL-MorphNet, a novel deep learning model designed to classify ALL cells from microscopic images. The model starts by using a convolutional stem block to capture small local details from the images. Next, a dynamic morphological block analyzes the images at multiple scales to capture both fine- and coarse-grained patterns. This step uses an attention mechanism to focus on the most important features. Subsequently, a fusion module concatenates these features without information loss. Finally, a hybrid transformer block incorporates long-range relationships into a comprehensive understanding of the image. We evaluated the model on the ALL-IDB2 dataset, achieving an accuracy of 94%, outperforming traditional CNNs, such as ResNet, VGG, and transformer-based models, which achieved 85-90% accuracy on the same dataset. Future studies will aim to improve the model practicality to a variety of datasets, multimodal inputs, and enhanced morphological processing. Additionally, we focused on optimising the framework for deployment in resource-constrained environments to ensure its applicability to real-world clinical workflows. This advancement contributes to enhancing early and precise ALL diagnostics and reducing healthcare disparities in both high-income and low-and-middle-income countries.

Keywords: *Acute Lymphoblastic Leukaemia (ALL); Microscopic Image Analysis; ALL-IDB2 Dataset; Deep Learning (DL); Convolutional Neural Networks (CNNs); Transformer-Based Models; Dynamic Morphological Features; Hybrid Convolutional Transformer Architecture; Cross-Attention Mechanism;*

I. INTRODUCTION

Acute lymphoblastic leukaemia (ALL) is a deadly disease caused by the uncontrolled growth of abnormal white blood cells, primarily affecting children age 1-4 years. It continues to be one of the leading causes of cancer-related deaths in this age group (Onciu, 2009) [2]. Recent US statistics show that ALL affects about 1.6 in every 100,000 people, with 1,441 deaths reported in 2022[1].

In high-income nations, the survival rates for ALL patients have increased remarkably to over 90%. However, in low-and middle-income areas, survival rates vary between 22% and 90%. Delays in diagnosis, financial barriers, and limited access to healthcare contribute to these disparities[2,3]. Diagnosis of ALL is challenging because

malignant cells do not possess distinct features compared to other types of leukaemia. Pathologists must rely on cutting-edge complex tests, such as identifying specific cell markers, genetic tests, and checking for leftover cancer cells after treatment. These tests are time-consuming and require expensive equipment, and sometimes the results can vary in accuracy[4,5]. Manual examination under a microscope also depends heavily on the skill of the observer, leading to human errors and challenging to make an accurate diagnosis without suitable equipment. This highlights the need for a solution that can automatically analyse images while considering both the small details and broader patterns.

In recent years, the rapid development of artificial intelligence (AI) has opened up new opportunities across various domains. Similarly, medical imaging tasks have seen significant progress with the development of deep learning architectures such as CNNs, AlexNet, DenseNet, ResNet, MobileNetv2, YOLO, Attention-Based CNNs, WBCs-Net and Alert-Net. In a systematic review conducted by Das et al., 2022, it was revealed that traditional and recent CNN models influenced various aspects of healthcare, such as enhanced diagnostic accuracy, early detection or timely intervention of diseases, reduction in human error and consistency in diagnosis, automation of routine tasks, augmented clinical workflows, integration with electronic health records (EHR), and expansion of medical knowledge[6]. In addition, research studies by Ahmed et al., 2023[7] state that AI enhances diagnostic accuracy, early detection of diseases, and reduction in cost and time.

Conversely, despite the advancements brought by CNNs and other machine learning models, several challenges and obstacles persist. These challenges affect the implementation and effectiveness of technologies, as well as healthcare systems. Some of the key challenges include observer bias, imbalanced datasets, and expensive computational demands [5], data privacy and security, inconsistent data, resistance to change, algorithm bias and generalisation issues, clinical validation and trust or regulatory hurdles, and ethical questions about the extent to which machines should be involved in clinical decision-making[6]. Furthermore, these approaches struggle to capture detailed features, such as irregular cellular shapes, alongside cancer progression patterns. These identified limitations highlight the need for additional research to address these challenges and make AI tools more reliable and efficient.

The findings motivated the development of ALL-MorphNet, a novel framework designed to improve ALL cell

classification using the ALL-IDB2 dataset. The model leverages an attention mechanism to accurately extract features that harmonise fine-grained details with overall contextual information. We utilised depth-wise convolutions to extract features at multiple scales and a transformer mechanism to identify long-range patterns within the image. This enhances the leukaemia cell classification performance, effectively addressing the challenges in ALL image processing. This study makes several important contributions, which are outlined below.

Innovative Architecture: The framework integrates depth-wise convolutions with an attention mechanism inspired by a transformer. Depth-wise convolutions are effective for capturing detailed morphological features, whereas transformers are designed to represent long-range dependencies. This hybrid method connects in-depth structural analysis with a broader contextual comprehension.

Performance Evaluation: The proposed approach outperformed the current state-of-the-art methods by delivering a 3.6% increase in accuracy compared to Vision Transformers and a 4.7% improvement over U-Nets [8]. These results highlight the accuracy, robustness, and high performance of our approach for classifying ALL.

Ablation Studies: By conducting systematic experiments, we validated the significance and relevance of each component in our novel framework. The importance of dynamic morphology processing and attention mechanisms has been identified as crucial for improving the performance. These studies indicate that every aspect of the proposed architecture contributes significantly to the success of the model.

Generalisability Validation: The reliability of the model was evaluated through multiple experimental trials. The aggregated results were used to evaluate the reliability of the model. This test highlights the framework's ability to effectively generalise to unknown data.

In conclusion, ALL-MorphNet enhances the classification accuracy and robustness by integrating localised structural details and global contextual information. This innovative approach not only improves risk assessment and enables earlier detection but is also essential for better patient outcomes. Ultimately, this study represents a significant advance in cancer research, helping to alleviate the impact of leukaemia and enhance diagnostic processes.

II. PROPOSED METHODOLOGY

The experiments were performed on a Windows 10 laptop using WSL 2 (Ubuntu 22.04). The hardware specifications included an Intel 6-core CPU (2.59GHz), 32GB RAM, 400GB storage, and NVIDIA RTX 2070A MAX-Q (6GB). This configuration demonstrates the feasibility of implementing deep learning on standard hardware, effectively balancing efficiency and accessibility. However, we also recognise the scalability challenges associated with large datasets.

A. Dataset Description

The model was evaluated using the ALL-IDB2 dataset, which is a subset of the larger ALL-IDB1 dataset. This dataset was carefully prepared by experts to evaluate algorithms designed for classifying or segmenting acute lymphoblastic leukemia (ALL) cells [9]. The dataset comprises 260 regions of interest (ROIs), each sized 257×257 pixels, cropped from peripheral blood smear images within the ALL-IDB1 dataset. Each image was centred on a single white blood cell,

ensuring that the dataset was effectively curated for in-depth analysis. This high-quality dataset serves as a dependable resource for researchers to assess the effectiveness of models in accurately classifying ALL cells. In summary, the accuracy of the findings relies heavily on the data quality.

Dataset Composition: The dataset had an equal distribution of 130 lymphoblasts (cancerous cells) and 130 normal lymphocytes. This balanced data distribution addresses the class imbalance issues in clinical datasets that support the model reliability assessment.

Image Specifications: The blood cell images have a resolution of 257×257 pixels and a 24-bit colour depth and were saved in TIF format. These high-resolution cell images guarantee the availability of fine- and coarse-grained features for precise classification, and the consistent image attributes make pre-processing easier.

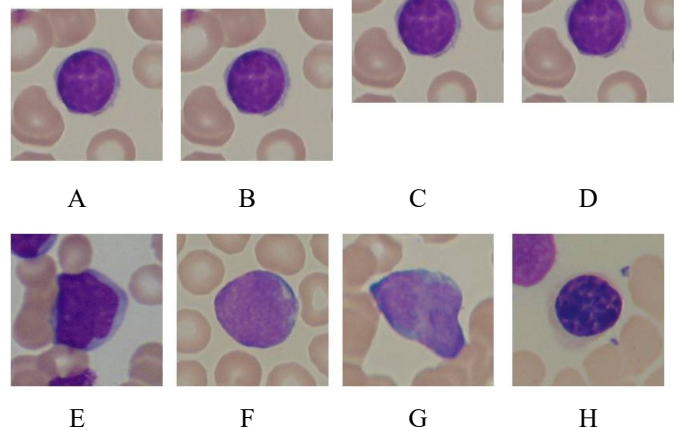


Figure 1. The ALL-IDB2 dataset for acute lymphoblastic leukemia

Figure 1 shows the normal cells (A-D) and the suspected lymphoblasts from individuals (E-H).

Image Acquisition: The Photographs of the blood smears were taken with a Canon PowerShot G5 camera and an optical microscope at 300x to 500x magnification. The ALL-IDB2 dataset was selected for several reasons.

Balanced Class Distribution: The dataset contains an equal number of lymphoblasts and normal lymphocytes. This parity addresses the issue of class imbalance, which often poses difficulties in training classification models.

Compact Size: The study was conducted in environments with standard hardware. With only 260 images, the limited size of the dataset makes it practical and manageable for use in these settings.

Public Availability: ALL-IDB2 is freely accessible to researchers, promoting reproducibility and benchmarking. This accessibility allows others to verify the findings and expand the research, encouraging collaboration and trust in the results.

Standardisation: Specialists carefully created the dataset using methods that ensured high data quality. Their efforts ensured that the dataset was both reliable and consistent, which is essential for meaningful analysis.

These characteristics of ALL-IDB2 are an important resource for enhancing research on the diagnosis of acute lymphoblastic leukaemia (ALL). By addressing issues such as class imbalance and availability, the dataset lays a solid foundation for creating improved diagnostic techniques.

Comparison to Other Datasets

C-NMC-2019, Raabin-Leukemia, and BCCD are important datasets because of their size and variety of samples. However, they face challenges such as class imbalance and high computational requirements. We chose ALL-IDB2 because of its controlled nature, which enabled us to effectively build and validate our architecture prior to tackling more complex tasks.

Limitations of ALL-IDB2

Sample Size: The dataset was relatively small, containing only 260 images. To mitigate this limitation, we applied data-augmentation techniques to increase the amount of available data.

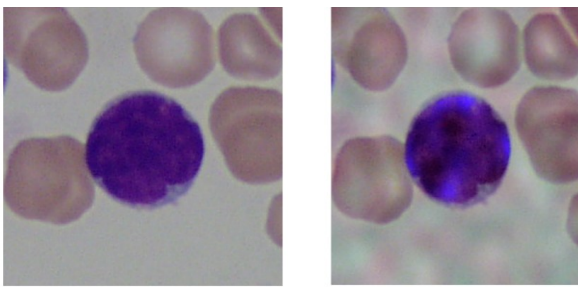
Demographic Diversity: The absence of donor demographic information raises concerns regarding the effectiveness of the dataset in representing various populations. Consequently, the model may have difficulty accommodating differences specific to certain groups.

Subtype Classification: The dataset does not provide information to differentiate between B-ALL and T-ALL, which restricts its potential for broader clinical use.

To address these challenges, we employed specific pre-processing and augmentation strategies. These measures improved the model's performance and its ability to generalise to various scenarios.

B. Image pre-processing and Augmentation

Photometric Augmentations: To handle variations in image quality, we applied a Gaussian blur using a 5×5 kernel. We adjusted the sigma value between 0.1 and 1.5 to create different degrees of blurring. This technique was employed to replicate the flaws typically found in microscopy images, such as those resulting from focus or lighting changes, and by incorporating this method to enhance the model's robustness against real-world imaging difficulties. This strategy ensured that the model learned to process imperfect images while maintaining accuracy and mirrored the types of variations frequently experienced during actual data collection. Figure 2 shows the original and augmented image.



(a) Original Image (b) Augmented Image

Figure 2. Impact of Wright-Giemsa Stain Perturbation on Microscopic Images

Furthermore, to address the differences in staining across microscope slides[10], we introduced a new module called Wright-Giemsa Stain Perturbation as part of our data augmentation process. This technique breaks down images into their fundamental staining components using a fixed matrix. Subsequently, a small amount of smooth noise was added to mimic real-world changes in stain intensity. We manage this process with two key settings: a perturbation factor of 0.5, which fine-tunes the level of variation, and a smoothing parameter ($\sigma=10.0$) that helps ensure that the changes appear natural.

As we continue to refine our approach, it is crucial to understand how these adjustments benefit the model. By exposing the model to a variety of staining patterns, our technique enables it to handle inconsistencies during slide preparation more effectively. This adaptability minimises overfitting while enhancing the model's ability to process new data. Figure 2 exemplifies this idea, displaying a side-by-side comparison of the original image and a modified version that introduces realistic stain variations while ensuring that key details remain intact.

Ultimately, this approach aligns with our objective of addressing real-world challenges, such as uneven data and image variability. By implementing these methods, we strive to create a more reliable and impactful diagnostic tool. Looking ahead, there is plenty of potential for future studies to delve deeper; researchers could experiment with further adjustments to the parameters or even apply this technique to explore other staining methods.

C. Architecture Formulation

A new deep-learning architecture called ALL-MorphNet. was designed to address the challenges of detecting leukaemia in blood smear images. The schematic representation of the image is shown in figure 3. These challenges include complex cell shapes, variations in staining, limited annotated data, and computational-resource constraints. The architecture combines dynamic morphological processing with a transformer-based-attention mechanism. It follows a process similar to that of a pathologist. First, the sample was preprocessed using the Convolutional Stem Block. Next, fine details are examined using the Dynamic Morphological Block. Then, it correlates the findings with the Context Aware Attention Block. Subsequently, the results are verified through multiple checks using the Hybrid Transformer Block. Finally, the diagnosis is made using the Classification Head. This design allows the model to capture both small-scale features and broader context. This improves the detection of cancerous cells. This approach bridges the gap between detailed analysis and overall understanding, making it highly effective for leukaemia detection.

$$f_{\theta}: \mathcal{X} \rightarrow \mathcal{Y}, \text{ where } \mathcal{X} \subset \mathbb{R}^{H \times W \times 3} \text{ (RGB images)} \text{ and } \mathcal{Y} = \{0,1\} \text{ (healthy/leukemic).}$$

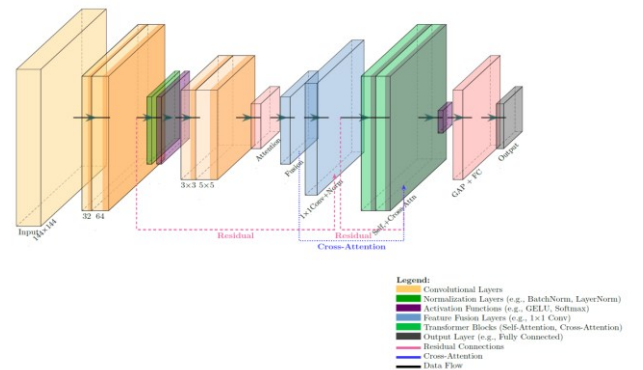


Figure 3. Schematic Representation of the ALL-MorphNet Architecture

The main components of this architecture in detail.

Convolutional Stem Block: The Convolutional Stem Block extracts initial features using two sequential 3×3 convolutions. The first convolution has 32 filters $\text{Conv}_{3 \times 3}^{32}$, and the second has 64 filters $\text{Conv}_{3 \times 3}^{64}$. After each convolution, batch normalization (BN_c) and GELU

activations are applied. This process helps address problems such as inconsistent staining, lighting conditions, and variations in cell shapes in acute lymphoblastic leukemia (ALL) microscopy images. It focuses on capturing key details, such as the boundaries between the nucleus and cytoplasm, the cell edges, and the irregular nuclei. These features are vital for distinguishing ALL blasts from normal lymphocytes. Additionally, the block gradually increased the number of channels from 32 to 64. This shift helps move from simple details, such as edges, to more complex patterns, such as irregular nuclei. Simultaneously, residual connections ensure that subtle features, such as visible nucleoli and hyperchromatic nuclei, are not lost.

By combining these techniques, the Convolutional Stem Block effectively prepares the data for further analysis. This ensures that both fine details and broader patterns are captured accurately.

$$X_{\text{stem}} = \text{GELU} \left(\text{BN}_{64} \left(\text{Conv}_{3 \times 3}^{64} \left(\text{GELU} \left(\text{BN}_{32} \left(\text{Conv}_{3 \times 3}^{32}(x) \right) \right) \right) \right) \right)$$

Dynamic Morphological Block: The Dynamic Morphological Block extracts features at multiple scales using depth-wise convolutions, spatial attention, and channel recalibration. Here is how it works, step by step:

Multiscale Feature Extraction: The block uses depth-wise convolutions with kernel sizes $k \in \{3, 5\}$. The smaller kernels (3×3) focused on fine details, such as nucleoli and nuclear lobulation. Larger kernels (5×5) capture broader patterns, such as cell clustering and irregular cell shapes. This approach helps to handle the variations seen in ALL blasts, such as irregular nuclei and cytoplasmic basophilia. By combining different kernel sizes, the block ensures that both small and large structures are captured effectively. This method ensures that the model can handle the complexity of acute lymphoblastic leukaemia (ALL) images while maintaining accuracy. It also prepares the data for the next stage of analysis by preserving important features.

$$\text{For each kernel size } k \in \{3, 5\}: F_k = \text{GELU} \left(\text{DWConv}_{k \times k}^C(X_{\text{stem}}) \right)$$

Spatial Attention: A 1×1 convolution with softmax weighting calculates scale-specific weights to focus on important features. This helps to manage variations in key features across different tasks. As a result, the model highlights critical pathology indicators, such as irregular nuclei, while reducing attention to less useful details.

$$A_{\text{spatial}}[k] = \text{Softmax} \left(\text{Conv}_{1 \times 1}(\text{GlobalAvgPool}(X)) \right)$$

Feature Fusion: The process combines weighted feature maps and then uses a 1×1 convolution to reduce their dimensions. This step ensured that both small- and large-scale structural information were preserved. By integrating fine- and coarse-grained patterns into a single representation, the model effectively captures essential pathological features.

$$F_{\text{fused}} = \text{Conv}_{1 \times 1} \left(\bigoplus_k [F_k \odot A_{\text{spatial}}[k]] \right)$$

Channel-wise recalibration: The bottleneck layers amplify important channels, whereas a sigmoid activation ensures smooth scaling. This addresses challenges such as computational overhead and feature prioritisation. The method focuses on key features, such as hyperchromatic

nuclei and cytoplasmic vacuolation, ensuring that they stand out.

$$a_{\text{channel}} = \sigma \left(\text{Conv}_{1 \times 1}^C \left(\text{GELU} \left(\text{Conv}_{1 \times 1}^{C/4} \left(\text{GlobalAvgPool}(F_{\text{fused}}) \right) \right) \right) \right)$$

Final Output: Recalibrated features are adjusted using a learnable parameter (γ) and regularised with dropout. This prevents overfitting and balances the feature importance. As a result, the model emphasizes critical pathological markers based on task requirements, ensuring accurate and reliable results.

$$M = \text{Dropout} (F_{\text{fused}} \cdot a_{\text{channel}} \cdot \gamma)$$

Fusion Layer: The Fusion Layer combines morphological features (M) with intermediate features from the stem block. It uses a 1×1 convolution and layer normalisation to achieve this. This layer also acts as a projection layer, aligning the dimensions required for the transformer blocks. It helps solve problems like information loss during feature transitions and inconsistent imaging conditions. By blending local structural details, such as irregular nuclei, with low-level spatial cues like edges, it ensures a strong and consistent representation of key pathological features

$$Z_{\text{fusion}} = \text{LayerNorm}(\text{Conv}_{1 \times 1}(M \oplus X_{\text{stem}}))$$

Hybrid Transformer Block: The Hybrid Transformer Block applies self-attention, cross-attention, and feed-forward networks (FFNs) with residual connections. These components help the model understand both the local and global relationships in the data.

Self-attention: Self-attention calculates attention scores by comparing queries (Q) and keys (K) and then applies these scores to values (V). This process links the relevant features across the images. For example, it connects irregular nuclear shapes in distant regions. This mechanism addresses the challenge of modelling long-range dependencies, such as abnormal clustering of cells. As a result, it provides a comprehensive view of cellular morphology.

$$\text{Attention} = \text{Softmax} \left(\frac{Q \cdot K^T}{\sqrt{d_k}} \right) \cdot V$$

Cross-attention: Cross-attention integrates morphological features (M) into the attention mechanism. It links localised structures, such as nuclear lobulation, to broader patterns, such as cell distribution. This approach overcomes the limitations of architectures such as U-Nets, which often struggle with long-range dependencies. By correlating local and global patterns, it helps differentiate ALL subtypes, improving diagnostic accuracy.

$$\text{Cross-Attention} = \text{Softmax} \left(\frac{Q \cdot K_{\text{morph}}^T}{\sqrt{d_k}} \right) \cdot Q_{\text{morph}} +$$

Feed-Forward Network (FFN): The FFN refines features using two-layer 1×1 convolutions with GELU activation. This helps to capture subtle patterns, such as nucleolar prominence and cytoplasmic granularity. Consequently, the model becomes better at learning detailed representations.

$$\text{FFN}(X) = \text{Conv}_{1 \times 1}^C \left(\text{GELU} \left(\text{Conv}_{1 \times 1}^{E \cdot R}(X) \right) \right)$$

Residual Connection: Residual connections add input features back to the output of each subcomponent, such as self-attention and FFN. This ensures a stable gradient flow and improves the stability of deep architectures. It preserves

low-level features while helping the model learn higher-level abstraction. Consequently, it maintains the fine-grained details required for pathology analysis, along with broader contextual information.

$$Z_{out} = Z_{in} + \text{FFN}(Z_{in} + \text{Self-Attention}(Z_{in}) + \text{Cross-Attention}(Z_{in}, M))$$

Classification: The Classification Head consolidates feature maps into a unified embedding using adaptive pooling, layer normalisation, and linear projection. This design addresses challenges such as varying input sizes and ensures consistent performance across datasets. It produces probabilistic outputs that effectively distinguish ALL blasts from normal lymphocytes, thereby enhancing diagnostic precision.

$$y = \text{Linear}(\text{LayerNorm}(\text{GlobalAvgPool}(Z_L)))$$

D. Key Innovations and Justifications

Vision transformers (ViTs): Vision transformers (ViTs) treat images as sequences of image patches. This approach removes spatial hierarchies, causing them to lose fine-grained details that are critical for tasks such as cell classification. Moreover, their reliance on global self-attention leads to high computational costs, with a complexity of $O(N^2)$ complexity for N patches. (Kolesnikov et al., 2020) [30].

Innovations: To address these issues, the Dynamic Morphology module captures multiscale structural patterns using depth-wise convolutions. This helps to retain the important spatial information that ViTs lack. Additionally, the grouped convolutions in the transformer blocks reduce the complexity to $O(N \cdot C)$. This makes the attention scalable while preserving spatial relationships.

U-Nets Limitations: U-Nets are excellent at extracting local features but struggle to model long-range dependency. Their skip connections can also propagate noise, which often harms performance when dealing with highly variable medical data (Ronneberger, 2017) [11].

Innovations: In model, transformer blocks integrate morphological features (M) with a global context through cross-attention. This allows localised structures to communicate with broader patterns of activity. Furthermore, the learnable scalar (γ) dynamically balances the morphological and contextual features, unlike the fixed skip connections used in U-Nets.

Fixed scaling assumes that all morphological features are equally important for all tasks. However, this assumption is unrealistic and limits the flexibility.

In contrast, ALL-MorphNet uses adaptive scaling for this purpose. It optimizes γ during training, mimicking how biological vision systems prioritize important cues. This makes the model more adaptable to different tasks and improves its ability to focus on the relevant details.

IV. RESULTS AND DISCUSSIONS

The model performance is assessed in classification tasks using various metrics. The findings indicate high levels of accuracy, precision, recall, and overall dependability.

Accuracy: The model demonstrated an accuracy rate of 94.03%, indicating that it accurately classified approximately 94 out of every 100 cases. In addition, the 95% confidence interval, which ranged from 88.36% to 99.70%, further corroborated the model's reliability within this specified

range. Figure 4 illustrates the performance of the model during training and validation across multiple epochs.

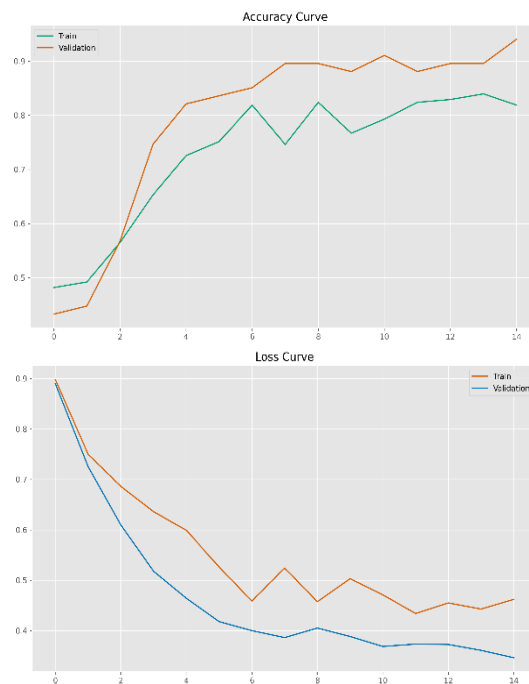


Figure 4. Training /Validation Performance Across Epochs

Precision: The model achieved a precision of 91.43%. This indicates that when the model predicted a positive outcome, it was accurate in more than 91% of instances. High precision is particularly critical in the automated classification of malignant cells, where false positives can have significant consequences. Figure 5 shows a confusion matrix that illustrates the classification performance of the model.

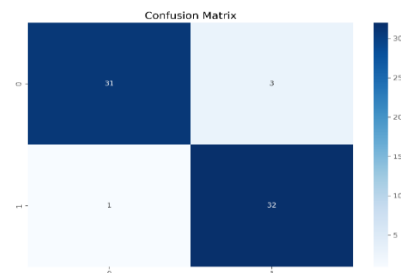


Figure 5. Confusion Matrix for Classification Performance

Recall: The recall score was 96.97%, indicating that the model effectively identified nearly all true-positive instances. A high recall is essential in the ALL classification context, where the omission of positive cases must be minimised. This metric reflects the sensitivity of the model in detecting significant patterns.

F1-Score: The F1-score achieved was 94.12%, indicating the model's proficiency in balancing precision and recall. This equilibrium is crucial for classification tasks that require both accuracy and sensitivity.

AUC Metric: To further evaluate the model's capacity to differentiate between classes, the area under the receiver operating characteristic curve (AUC-ROC) was calculated as 98.84%. Similarly, the area under the precision-recall curve

(AUC-PR) was 98.88%. These values underscore the exceptional proficiency of the model in distinguishing between positive and negative instances. Figure 6 shows the ROC and precision-recall curves for the model.

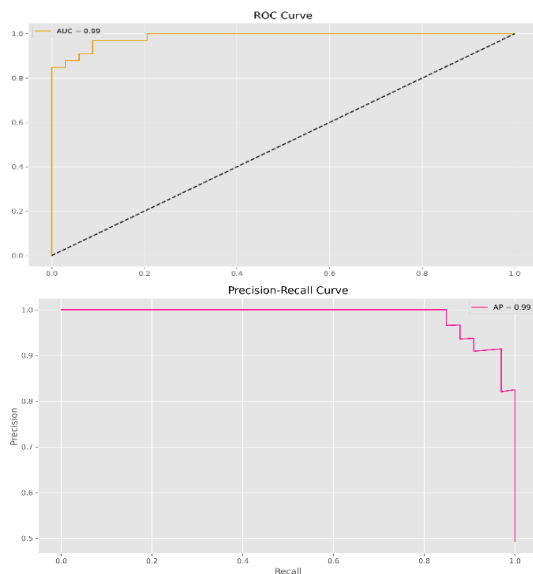


Figure 6. Receiver Operating Characteristic (ROC) and Precision-Recall Curves

Error Analysis: The normalised confusion matrix revealed balanced error rates. Figure 7 shows the normalised confusion matrix for the classification results.

- The true positive rate, or sensitivity, indicated that 97% of cancerous cells were accurately classified.
- The true-negative rate, also known as specificity, indicates that 91% of normal cells were accurately classified.
- The misclassification rates were relatively low, with false negatives at 3% and false positives at 9%.

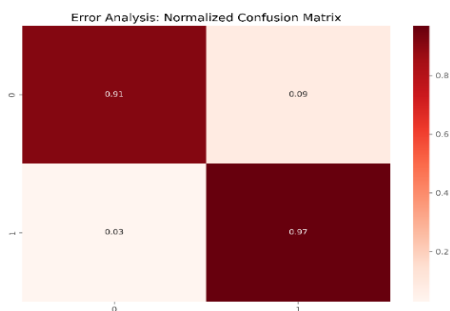


Figure 7. Normalized Confusion Matrix

Efficiency Metrics: The model exhibited a favourable computational profile, requiring a modest 3.71 Giga multiply accumulate floating point operations (GMac FLOPs) and employing 630.32 thousand parameters. This resulted in an average inference latency of 4.89 ms, suggesting the model's viability for real-time deployment, where minimal perceptible delays are crucial.

Statistical Significance: We tested the results using a binomial test, which gave a p-value of 5.53×10^{-15} . This low p-value provides substantial evidence that the observed accuracy did not occur by chance.

ALL-MorphNet demonstrated remarkable performance, achieving an accuracy of 94.03%, precision of 91.43%, recall of 96.97%, and F1-score of 94.12%. These metrics indicate that the model effectively balances the precision and recall while maintaining a superior performance. Furthermore, its average inference time of merely 4.89 ms emphasises its high efficiency, rendering it suitable for real-time clinical application.

A. Comparison with State-of-the-Art Models

To better understand these results, the proposed model is compared with recent approaches using similar datasets.

Study/Model	Models Used	Metrics	Comparison Summary
Baseline Models	DenseNet121, ResNet50, MobileNet	DenseNet121: Accuracy – 90.4%, AUC – 93.8%, Recall – 90.15% ResNet50: Accuracy – 90.3%, AUC – 92.35% MobileNet: Accuracy – 90.4%	The proposed model outperformed these in terms of accuracy, AUC-ROC, and recall. It eliminates the need for heavy pre-processing (e.g., artefact removal) via attention mechanisms
Deshpande et al. (2024) [16]	VGG16, Xception, DenseNet	Xception + XGBoost: Accuracy, Precision, Recall, F1-score – 91% each DenseNet + XGBoost: Avg Accuracy – 88%	Our model performed better, particularly in recall and AUC-ROC. Deshpande et al. noted generalization issues due to dataset limitations, which we addressed.
Baluabid et al. (2025) [17]	SVM, ResNet50, Hybrid (ResNet50 + SVM)	Hybrid (ResNet50 + SVM): Accuracy – 95%, Precision – 92%, Recall – 97%, F1-score – 94% ResNet50: Accuracy – 89% DenseNet121: Accuracy – 89%, F1-score – 78%	While the hybrid model showed slightly better accuracy, our model demonstrated superior AUC and scalability by incorporating learnable scaling (γ), reducing manual dependencies

V. CONCLUSION

The ALL-MorphNet architecture addresses significant challenges in leukaemia detection by integrating dynamic morphological processing and transformer-based attention mechanisms. This hybrid methodology effectively captures both small-scale structural details and broader contextual relationships, thereby achieving state-of-the-art performance on the ALL-IDB2 Dataset. The model achieved an accuracy of 94.03%, precision of 91.43%, recall of 96.97%, and F1-score of 94.12%. The dependence on a single dataset, ALL-IDB2, is a key limitation of the proposed architecture in this study. Although this dataset provides a solid starting point, its limited diversity in imaging conditions, staining techniques, and patient demographics may reduce the model's ability to perform well in real-world clinical settings. Variations in microscope resolution, lighting, and sample preparation methods can affect the performance of the model when applied to unseen data from different populations or institutions. The model uses only one dataset, restricting its generalizability. Additionally, the lack of detailed demographic information and subtype classifications (e.g. B-ALL vs. T-ALL) limits its use in broader clinical applications. The proposed model has the potential to significantly improve leukemia diagnosis by reducing errors, enhancing treatment outcomes, and easing the burden on healthcare systems. Its deployment in low-resource settings could bridge gaps in diagnostic capabilities, providing equitable access to timely and accurate healthcare. Furthermore, the integration of efficient computational methods ensures scalability, making it a practical solution for real-world clinical applications.

REFERENCES

- [1] SEER*Explorer: An interactive website for SEER cancer statistics [Internet]. Surveillance Research Program, National Cancer Institute; 2024 Apr 17. [updated: 2024 Nov 5; cited 2025 Mar 10]. Available from: <https://seer.cancer.gov/statistics-network/explorer/>. Data source(s): SEER Incidence Data, Nov 2023 Submission (1975-2021), SEER 22 registries.
- [2] Malard F, Mohty M. Acute lymphoblastic leukaemia. *Lancet*. 2020;395(10230):1146-62. [https://doi.org/10.1016/S0140-6736\(19\)33018-1](https://doi.org/10.1016/S0140-6736(19)33018-1) PMID:32247396 DOI: [https://doi.org/10.1016/S0140-6736\(19\)33018-1](https://doi.org/10.1016/S0140-6736(19)33018-1)
- [3] Duffy, Caitlyn, Dylan E. Graetz, Arturo M. Zapata Lopez, Angela K. Carrillo, Godwin Job, Yichen Chen, Meenakshi Devidas et al. "Retrospective analysis of outcomes for pediatric acute lymphoblastic leukemia in South American centers." *Frontiers in Oncology* 13 (2023): 1254233.
- [4] Pui CH, Evans WE. Acute lymphoblastic leukemia. *N Engl J Med*. 1998 Aug 27;339(9):605-15. doi: 10.1056/NEJM199808273390907. PMID: 9718381.
- [5] 01Mustaqim, T., Suciati, N., & Fatichah, C. (2023). Deep Learning for the Detection of Acute Lymphoblastic Leukemia Subtypes on Microscopic Images: A Systematic Literature Review. *IEEE Access*, 11, 16108–16127. <https://doi.org/10.1109/access.2023.3245128>
- [6] Das, P. K., Abraham, A., Panda, R., Meher, S., & A, D. V. (2022). A Systematic Review on Recent Advancements in Deep and Machine Learning Based Detection and Classification of Acute Lymphoblastic Leukemia. *IEEE Access*, 10, 81741–81763. <https://doi.org/10.1109/access.2022.3196037>
- [7] 02Ahmed, I. A., Senan, E. M., Shatnawi, H. S. A., Alkhraisha, Z. M., & Al-Azzam, M. M. A. (2023). Hybrid Techniques for the Diagnosis of Acute Lymphoblastic Leukemia Based on Fusion of CNN Features. *Diagnostics*, 13(6), 1026. <https://doi.org/10.3390/diagnostics13061026>
- [8] Baluabid, R., Alnasri, H., Alowaybidi, R., Hafiz, R., Alsini, A., & Alharbi, M. (2025). Detecting Acute Lymphocytic Leukemia in Individual Blood Cell Smear Images. *Engineering, Technology & Applied Science Research*, 15(1), 19167–19173. <https://doi.org/10.48084/etasr.9123>
- [9] 03Labati, R. D., Piuri, V., & Scotti, F. (2011, Sep 1). All-IDB: The acute lymphoblastic leukemia image database for image processing. <https://doi.org/10.1109/icip.2011.6115881>
- [10] Rayappana, L., & Parameswarib, R. (2024). Statistical and Algorithmic Color Normalization and Stain Separation Methods: A Brief Study (pp. 146–152). *crc*. <https://doi.org/10.1201/9781003430452-21>